Validation homework week6

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mrgsolve model

Load your mrgsolve model

```
# # Please load your model
# # Capture the individual predictions as a variable named `IPRED`.
# mod <-
mod <- mread("hwwk6.cpp", project=here("wk07", "mrg-model"))
```

```
Building hwwk6_cpp ... done.
```

see model

```
see(mod)
```

```
Model file:  hwwk6.cpp

[ prob ]
  This is my homework assignment for week 6

[ cmt ] depot cent

[ param ]
  WT=70  // weight in kg
  AGE=35 // age in years

[ nmext ]
  path = "../nm-model/hwwk6/hwwk6.ext"
  root = "cppfile"

[ pk ]
  // Typical values of PK parameters
  double TVCL   = THETA(1);
  double TVV    = THETA(2);
  double TVKA   = THETA(3);
  double TVD1   = THETA(4);

  // PK covariates
  double CL_AGE = pow((AGE/35.0), THETA(5));
  double CL_WT  = pow((WT/70), THETA(6));
  double V_WT   = pow((WT/70), THETA(7));

  double CLCOV = CL_AGE*CL_WT;
  double VCOV  = V_WT;

  // PK parameters
  double CL = TVCL*CLCOV*exp(ETA(1));
  double V  = TVV*VCOV*exp(ETA(2));
```

```
  double KA = TVKA;
  double D1 = TVD1*exp(ETA(3));

  double S2 = V;

  D_depot = D1;

[ ode ]
  dxdt_depot = -KA*depot;
  dxdt_cent = KA*depot - (CL/V)*cent;

[ error ]
  double IPRED = cent/S2;
  double Y = IPRED * (1+EPS(1));

[ capture ] CL V KA D1 IPRED Y WT AGE
```

Prepare input

Load your NONMEM input dataset

```
csv <- read_csv(here("wk07/data/nmdat1.csv"), na = ".")
```

```
Rows: 467 Columns: 14
-- Column specification --------------------------------------------------------
Delimiter: ","
dbl (13): NUM, ID, TIME, AMT, RATE, EVID, MDV, CMT, DV, BLQ, WT, AGE, DOSE
lgl  (1): C

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
head(csv, n = 3)
```

```
# A tibble: 3 x 14
  C       NUM    ID  TIME   AMT  RATE  EVID   MDV   CMT    DV   BLQ    WT   AGE
  <lgl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
```

```
1 NA        1     1   0    1000    -2     1     1     1  0        2   116    15
2 NA        2     1   0.1     0     0     0     0     2  0.25     1   116    15
3 NA        3     1   0.2     0     0     0     0     2  1.35     0   116    15
# i 1 more variable: DOSE <dbl>
```

Load NONMEM output table contains PRED

The PRED and IPRED column in this table is generated by NONMEM. Let's rename it as NMPRED and NMIPRED to avoid confusion.

```
tab <- read_table(here("wk07/nm-model/hwwk6/hwwk6.tab"),
                  skip = 1) %>% rename(NMPRED=PRED, NMIPRED=IPRED)
```

```
-- Column specification --------------------------------------------------------
cols(
  NUM = col_double(),
  IPRED = col_double(),
  NPDE = col_double(),
  CWRES = col_double(),
  DV = col_double(),
  PRED = col_double(),
  RES = col_double(),
  WRES = col_double()
)
```

```
head(tab, n = 3)
```

```
# A tibble: 3 x 8
    NUM NMIPRED   NPDE  CWRES    DV NMPRED    RES   WRES
  <dbl>   <dbl>  <dbl>  <dbl> <dbl>  <dbl>  <dbl>  <dbl>
1     1    0     0      0      0      0     0      0
2     3    1.36 -0.332 -0.234  1.35   1.83 -0.476 -0.162
3     4    6.63  0.210  0.440  7.7    8.86 -1.16   0.262
```

Join NONMEM input dataset and output table

The NUM column provides a key to join these two

```r
data <- left_join(tab, csv, by = "NUM")
head(data, n = 3)
```

```
# A tibble: 3 x 21
    NUM NMIPRED   NPDE  CWRES  DV.x NMPRED    RES   WRES C        ID  TIME   AMT
  <dbl>   <dbl>  <dbl>  <dbl> <dbl>  <dbl>  <dbl>  <dbl> <lgl> <dbl> <dbl> <dbl>
1     1    0     0      0      0      0      0      0     NA        1  0     1000
2     3    1.36 -0.332 -0.234  1.35   1.83 -0.476 -0.162 NA        1  0.2      0
3     4    6.63  0.210  0.440  7.7    8.86 -1.16   0.262 NA        1  0.45     0
# i 9 more variables: RATE <dbl>, EVID <dbl>, MDV <dbl>, CMT <dbl>, DV.y <dbl>,
#   BLQ <dbl>, WT <dbl>, AGE <dbl>, DOSE <dbl>
```

Validation using PRED

Simulate PRED using mrgsolve

```r
out <-
  mod %>%
  # Remove all random effects, so simulated IPRED=PRED
  zero_re() %>%
  # Take observation records only
  obsonly() %>%
  mrgsim(data = data,
         # Recover `NMPRED` for validation
         recover = "NMPRED",
         # Specify the output format as dataframe (df)
         output = "df",
         # Limit the number of output digits to 5 (what `NONMEM` is giving).
         digits = 5) %>%
  rename(MRGPRED=IPRED) # Rename mrgsolve `PRED` as `MRGPRED`

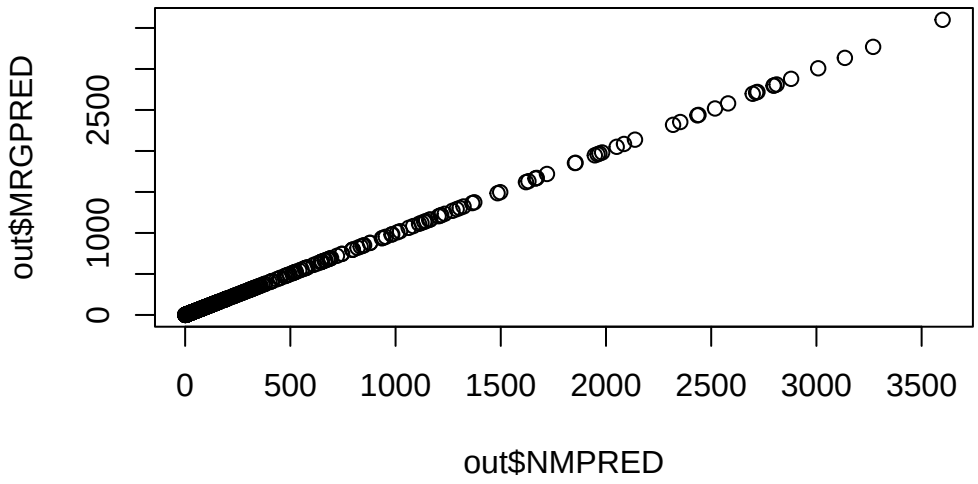
head(out, n = 3)
```

```
  ID TIME  depot     cent     CL      V      KA      D1 MRGPRED       Y  WT AGE
1  1 0.20 421.78   8.4192 1.4234 4.6116 0.20239 0.46471  1.8257  1.8257 116  15
2  1 0.45 925.56  40.8590 1.4234 4.6116 0.20239 0.46471  8.8600  8.8600 116  15
3  1 1.04 849.52 132.3400 1.4234 4.6116 0.20239 0.46471 28.6980 28.6980 116  15
   NMPRED
1  1.8257
```

```
2  8.8601
3 28.6980
```

Graphical check PRED by NONMEM versus mrgsolve

```
plot(out$NMPRED, out$MRGPRED)
```



Numerical check PRED by NONMEM versus mrgsolve

```
summary(100*(out$NMPRED - out$MRGPRED)/out$NMPRED)
```

```
      Min.    1st Qu.     Median       Mean    3rd Qu.       Max.
-5.103e-03  0.000e+00  0.000e+00 -5.958e-05  0.000e+00  3.198e-03
```

Validation using IPRED

Read in an additional table that contain the post-hoc ETA values

Validation using IPRED values needs an additional table which has post-hoc ETA values.

```
etas <- read_table(here("wk07/nm-model/hwwk6/hwwk6par.tab"),
                   skip = 1) %>%
  dplyr::select(matches("NUM|ETA"))
```

```
-- Column specification --------------------------------------------------------
cols(
  NUM = col_double(),
  CL = col_double(),
  V = col_double(),
  KA = col_double(),
  D1 = col_double(),
  ETA1 = col_double(),
  ETA2 = col_double(),
  ETA3 = col_double()
)
```

Join the additional dataset with post-hoc ETA values with the data created earlier. Again, the NUM column provides a key to join these two

```
data <- left_join(data, etas, by = "NUM")
```

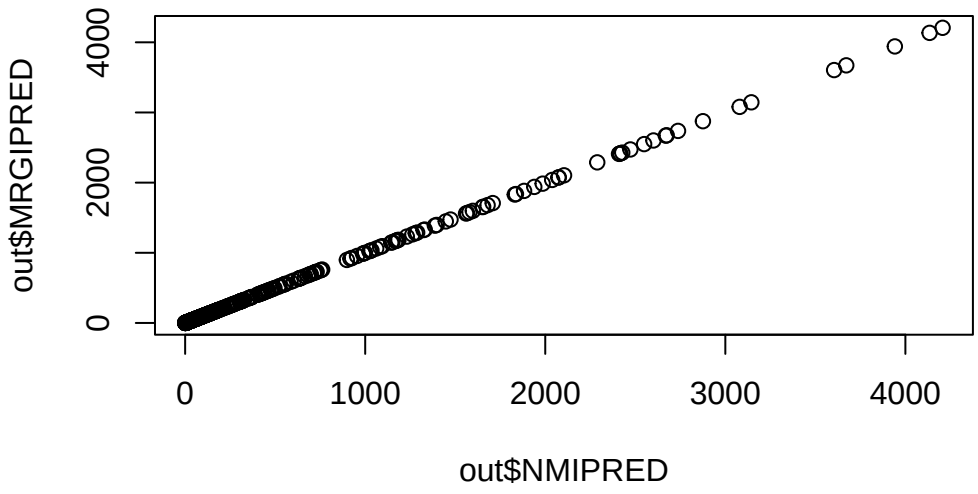
Simulate IPRED using mrgsolve

```
out <- mrgsim(
  mod,
  data,
  # Specify the source of ETA values coming from data
  etasrc = "data.all",
  # Take observation records only
  obsonly = TRUE,
  # Recover `NMPRED` for validation
  recover = "NMIPRED, NMPRED",
```

```
  # Specify the output format as dataframe (df)
  output = "df",
  # Limit the number of output digits to 5 (what `NONMEM` is giving).
  digits = 5
) %>% rename(MRGIPRED=IPRED)
```

Graphical check IPRED by NONMEM versus mrgsolve

```
plot(out$NMIPRED, out$MRGIPRED)
```



Numerical check PRED by NONMEM versus mrgsolve

```
summary(100*(out$NMIPRED-out$MRGIPRED)/out$NMIPRED)
```

```
      Min.    1st Qu.     Median       Mean    3rd Qu.       Max.
-3.685e-03  0.000e+00  0.000e+00 -2.387e-05  0.000e+00  6.284e-03
```